Substitution

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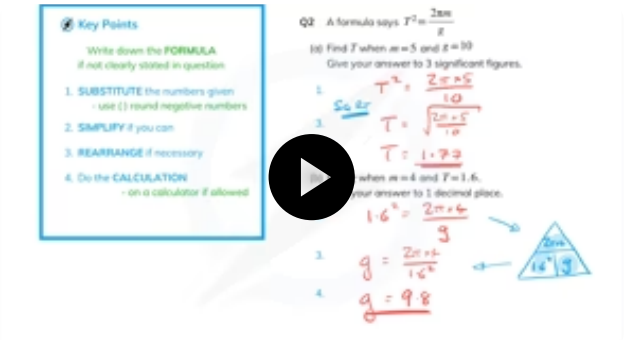

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Substitution

What is substitution?

- **Substitution** means **replacing** a letter (variable) in a formula with a given **number**

How do I substitute numbers into a formula?

- Write down the **formula**
- **Replace** (substitute) the letters in the formula with the given numbers
  - If substituting in a **negative number**, it is important to put **brackets** around it
    - For example, (-3)
- **Simplify** any numerical calculations
- **Calculate** the final value
- Sometimes the result is an **equation** which you can then **solve**

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

Examiner Tips and Tricks

Be careful with negatives and powers. For example, if $x = -4$ then $x^2 = (-4)^2 = 16$.

Worked Example

(a) Given that $x = 2$ and $y = -4$, find the value of the expression:

(i) $2x(x + 3y)$

(ii) $y^2 + y^3$

Answer:

(i)

Substitute the numbers given
Use brackets () around negative numbers

$$2 \times 2 \times (2 + 3 \times (-4))$$

Complete the calculation
Show every step of working, following the order of operations correctly

$$= 2 \times 2 \times (2 - 12)$$
$$= 2 \times 2 \times (-10)$$
$$= 4 \times (-10)$$

-40

(ii)

Substitute the numbers given
Use brackets () around negative numbers

$$(-4)^2 + (-4)^3$$

Work out the powers remembering to include the brackets

$$16 + (-64)$$

Complete the calculation

$$16 - 64$$

-48

(b) The formula $P = 2l + 2w$ is used to find the perimeter, P, of a rectangle of length l and width w.

Given that the rectangle has a perimeter of 20 cm and a width of 4 cm, find its length.

Answer:

Substitute the values you are given for P and w into the formula

$$20 = 2 \times l + 2 \times 4$$

Simplify

$$20 = 2l + 8$$

$12 = 2l$

Divide both sides by 2

$l = 6 \text{ cm}$

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